

Internal Benchmark Methodology for FUNGUS-SV

Assembly-Based Truth Set Construction for Non-Model Organisms

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Pipeline: FUNGUS-SV v0.5.1

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1. Background

Structural variant (SV) detection in non-model organisms lacks the gold-standard benchmark sets available for human genomes (GIAB, HGSVC, SMaHT). Without a truth set, it is impossible to measure precision, recall, or false discovery rates for any SV caller. This document describes a computational methodology for constructing an internal truth set using de novo genome assembly, enabling rigorous validation of SV detection pipelines in any organism.

The methodology is based on approaches used by: - **Kronenberg et al. (2025)** *Nature Methods* " Platinum Pedigree truth set construction - **Zhang et al. (2025)** *bioRxiv* " SMaHT mosaic SV benchmark - **Ebert et al. (2021)** *Science* " HGSVC assembly-based SV discovery - **Chen et al. (2023)** *Nature Communications* " DeBreak assembly validation

2. Principle

The core principle is: **SVs detected by comparing a de novo assembly to a reference genome are independent of read-alignment artifacts that affect read-based SV callers.** When both an assembly-based approach and a read-based approach (FUNGUS-SV) agree on an SV, the probability that it is a true variant is substantially higher than either method alone.

Why Assembly-Based SVs Are More Reliable

Evidence Type	Source	Artifacts
Read-alignment SVs	Mapped reads (BAM)	Mapping errors, chimeric alignments, repeat collapse
Assembly-based SVs	Contigs from de novo assembly	Assembly errors (less frequent with HiFi)
Concordant SVs	Both agree	Minimal " orthogonal methods confirm each other

The assembly-based approach has been validated in human genomics, where assembly-based SV calls from hifiasm + PAV achieve >90% concordance with GIAB truth sets (Ebert et al. 2021).

3. Methodology

3.1 Required Data

Item	Format	Purpose
PacBio HiFi reads	FASTQ (.fastq.gz)	From the isolate to be analyzed
Reference genome	FASTA (.fasta)	Different strain/species for comparison

3.2 Step 1: De Novo Genome Assembly

Assemble the HiFi reads into a haplotype-resolved genome assembly using hifiasm:

```
```bash # Install hifiasm conda create -n assembly -c bioconda hifiasm -y conda activate assembly
```

### Run assembly (haploid mode for fungi)

```
hifiasm -o isolate_asm -t 16 "primary isolate_reads.fastq.gz"
```

### Convert to FASTA

awk 'â€¢~/^S/{print â€¢œ>â€¢\$2;print \$3}'â€¢™ isolate\_asm.p\_ctg.gfa > isolate\_asm.p\_ctg.fasta Expected output: A haploid assembly with contig N50 > 1 Mb for fungal genomes at >50A— HiFi coverage.

### Quality control:

## Check assembly statistics

```
quast isolate_asm.p_ctg.fasta -r reference.fasta -o quast_results/ 3.3 Step 2: Assembly-Based SV Calling Compare the
assembled genome to the reference using SVIM-asm in haploid mode: # Align assembly to reference minimap2 -x asm5
-t 8 "cs reference.fasta isolate_asm.p_ctg.fasta
| samtools sort -@ 4 -o asm_to_ref.sorted.bam - samtools index asm_to_ref.sorted.bam
```

## Call SVs from assembly comparison (haploid mode)

```
svim-asm haploid --min_sv_size 50 asm_to_ref.sorted.bam reference.fasta
results/asm svsv/ Why SVIM-asm:
```

Designed for both haploid and diploid assemblies (Heller & Vingron 2020)

Validated on human HGSC data with high precision

## Reports DEL, INS, INV, DUP, and complex SVs

### 3.4 Step 3: Read-Based SV Detection (FUNGUS-SV) Run FUNGUS-SV on the same HiFi reads aligned to the same reference:

## Align reads to reference

```
minimap2 -t 8 -ax map-hifi -R @RG:isolate:sampleâ€™™
reference.fasta isolate_reads.fastq.gz
| samtools sort -@ 4 -o reads.to.ref.sorted.bam - samtools index reads.to.ref.sorted.bam
```

## Run FUNGUS-SV ICB consensus

```
python fungus_sv/core/icb.py
 --bam reads_to_ref.sorted.bam
 --reference reference.fasta
 --output results/icb/
 --callers sniffles2 cutesv svim
 --min-callers 2
```

## Run FUNGUS-SV validation

```
python -m valid_sv.run_validation
 --consensus-vcf results/icb/consensus_svsv.vcf
 --bam reads_to_ref.sorted.bam
 --reference reference.fasta
 --fastq isolate_reads.fastq.gz
 --output results/validation/ 3.5 Step 4: Benchmark Comparison Compare FUNGUS-SV results to the assembly-based
truth set using Truvari: # Compare ICB consensus to assembly SVs truvari bench
-b results/asm_svsv/variants.vcf
-c results/icb/consensus_svsv.vcf
 --reference reference.fasta
-o benchmark_results/
 --pctseq 0 --pctsize 0.7 --refdist 500
```

## Generate summary metrics

truvari summarize --input benchmark\_results/ --output summary.txt

### 3.6 Step 5: Stratified Performance Analysis

Compute precision, recall, and F1 score stratified by:

Stratification Why SV type (DEL, INS, INV, DUP) Different types have different detection difficulty SV size (50-100 bp, 100-500 bp, 500-5000 bp, >5 kb) Size-dependent accuracy T-score tier (HIGH  $\geq 0.6$ , MEDIUM 0.4-0.6, WEAK <0.4) Validates T-score calibration Repeat context (in repeat vs.  $\hat{A}$  unique) Repeat regions are challenging Number of supporting ICB callers (2 vs.  $\hat{A}$  3) Validates ICB consensus 4. Expected Outcomes 4.1 Performance Targets (Based on Human Benchmarks) Metric Target Source ICB Precision >80% SV-JIM (Todd et al.  $\hat{A}$  2025): 3-caller consensus ICB Recall (DEL) >60% Liu et al.  $\hat{A}$  (2024): best callers on PacBio ICB Recall (INS) >40% Liu et al.  $\hat{A}$  (2024): insertions are harder T  $\geq 0.6$  FDR <15% SV-MeCa (Nkouamedjo et al.  $\hat{A}$  2025) Ablation: multi-layer > single-layer  $\hat{\geq} 15\%$

improvement Expected from triangulation theory 4.2 T-Score Calibration The benchmark enables mapping T-score thresholds to empirical false discovery rates:

T-Score Expected FDR Action  $\hat{\pi} \leq 0.8$  <5% Suitable for functional follow-up  $\hat{\pi} \leq 0.6$  <15% Prioritized candidate  $\hat{\pi} \leq 0.4$  <35% Exploratory only  $<0.4$  >50% Likely false positive These thresholds are estimates pending empirical calibration.

5. Limitations Assembly errors can create false SVs. hifiasm assemblies of HiFi data have high accuracy (QV > 50), but errors in repetitive regions can introduce false SVs that will be misclassified as true positives.

Assembly-based callers miss complex SVs. SVIM-asm does not detect translocations or complex rearrangements. The truth set will be incomplete.

This is a silver standard, not gold. Unlike GIAB (which uses multiple technologies, pedigree information, and manual curation), this approach uses a single technology and automated methods.

Validated only for haploid genomes. Diploid or polyploid genomes require phasing before assembly, which introduces additional complexity.

6. Validation Against Published Methods To ensure the benchmark is reliable, cross-validate against orthogonal approaches:

Method Tool Comparison Different assembly-based caller PAV Compare to SVIM-asm results Different assembler Flye Compare to hifiasm assembly Read-only validation SVvalidation (Zheng 2024) Independent breakpoint confirmation High concordance (>80%) between methods increases confidence in the truth set.

7. Software Requirements Tool Version Installation hifiasm  $\hat{\pi} \leq 0.19$  conda install -c bioconda hifiasm minimap2  $\hat{\pi} \leq 2.30$  conda install -c bioconda minimap2 samtools  $\hat{\pi} \leq 1.17$  conda install -c bioconda samtools svim-asm  $\hat{\pi} \leq 1.0$  conda install -c bioconda svim-asm Truvari  $\hat{\pi} \leq 5.0$  conda install -c bioconda truvari quast  $\hat{\pi} \leq 5.0$  conda install -c bioconda quast
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This methodology is part of the FUNGUS-SV pipeline documentation. The pipeline is available at <https://github.com/keltonjenkovguimaraes-alt/fungus-sv>